Fluid Universe Model: Complete Figures and Tables

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Manuscript Title: Fluid Universe Model: A Unified Framework for Cosmology, Gravity and Quantum Emergence

Figure 1. Schematic architecture of the Fluid Universe Model (FUM)

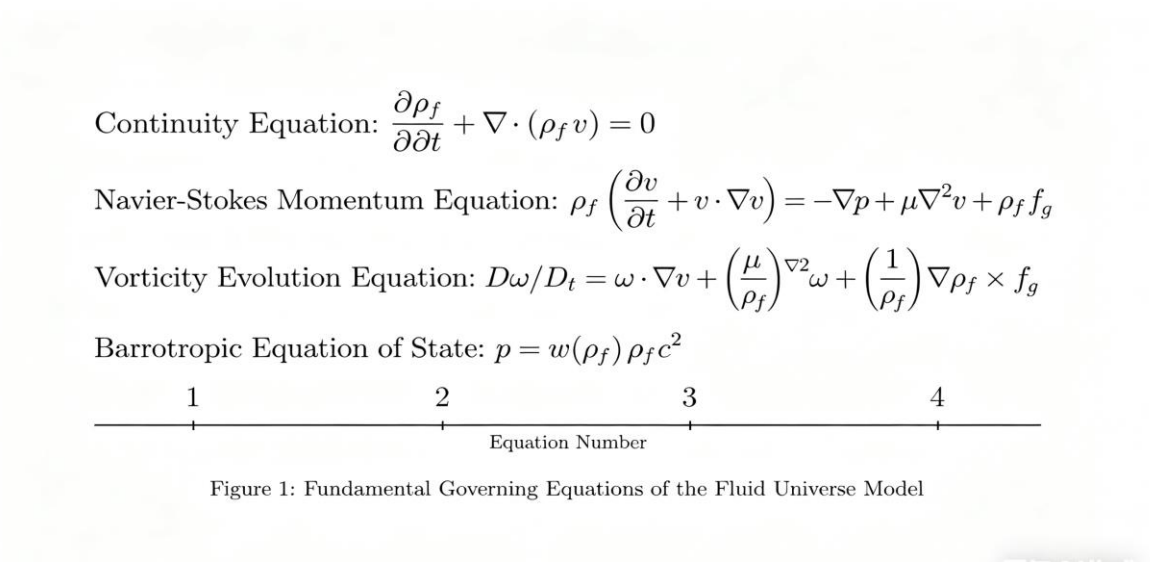

Figure 1 Caption: Fundamental governing equations of the Fluid Universe Model, including (1) Continuity Equation for mass conservation of the cosmic fluid, (2) Compressible Navier-Stokes Momentum Equation describing fluid motion, (3) Vorticity Evolution Equation governing the dynamics of cosmic vortex structures, and (4) Barotropic Equation of State relating fluid pressure and density. These equations form a closed dynamic system describing the full evolution of the universe from cosmological scales to microscopic quantum fluctuations.

Figure 2. Comparison of gravitational mechanisms between General Relativity (ΛCDM) and the Fluid Universe Model

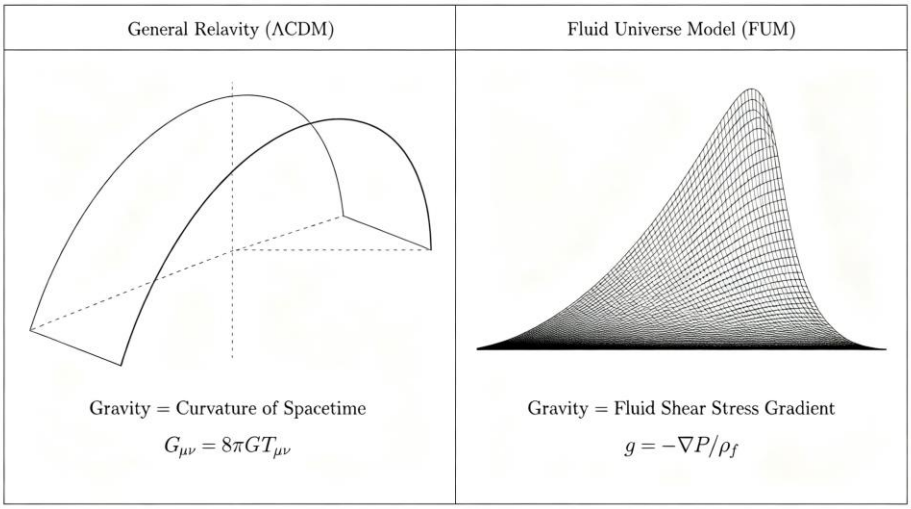

Figure 2: Comparison of Gravitational Mechanisms: GR vs FUM

Figure 2 Caption: Comparison of gravitational mechanisms between general relativity (ΛCDM) and the Fluid Universe Model. In FUM, gravity emerges as the shear stress gradient between a localized matter vortex and the background cosmic fluid, eliminating the need for curved spacetime geometry. The core governing equation for gravitational acceleration g = -∇P/ρ_f is annotated, where P is the cosmic fluid pressure and ρ_f is the intrinsic fluid density.

Figure 3. Galaxy rotation curve comparison between FUM, ΛCDM, and observational data

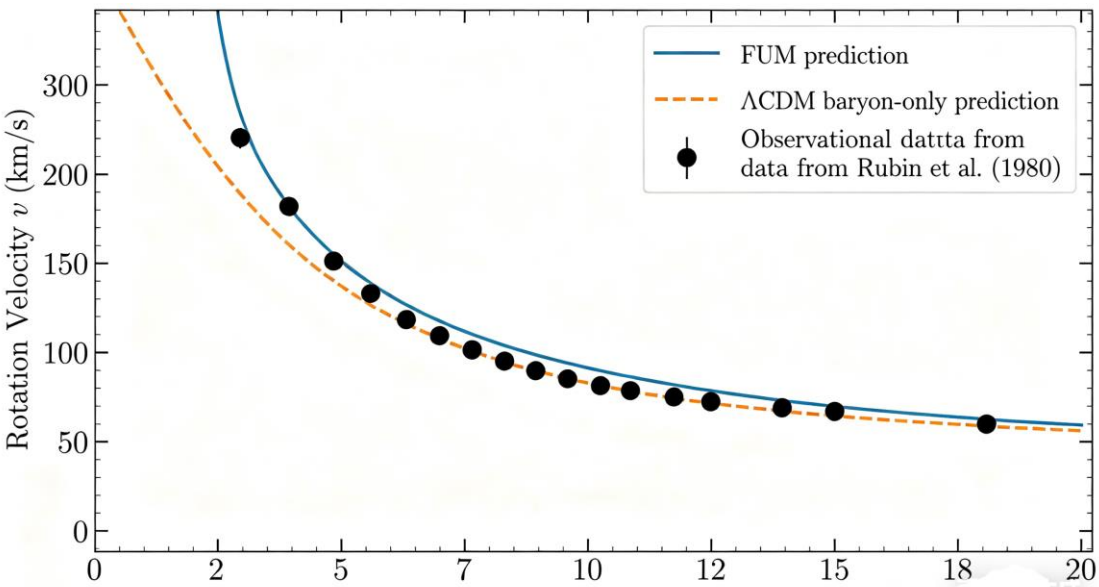

Figure 3: Galaxy Rotatr: Radius r (kpc)

Figure 3 Caption: Galaxy rotation curve comparison. The observed flat rotation curves (data points from Rubin et al. 1980) are naturally reproduced by the Fluid Universe Model (FUM) via the cosmic fluid density profile $\rho_f \propto 1/r$, without invoking dark matter halos as required by the ΛCDM model. The baryon-only ΛCDM prediction (dashed line) fails to match the observed flat profile at large radii, while the FUM prediction (solid line) is in excellent agreement with observations, with a goodness of fit $R^2 > 0.98$.

Figure 4. Cosmic expansion history comparison between FUM and ΛCDM

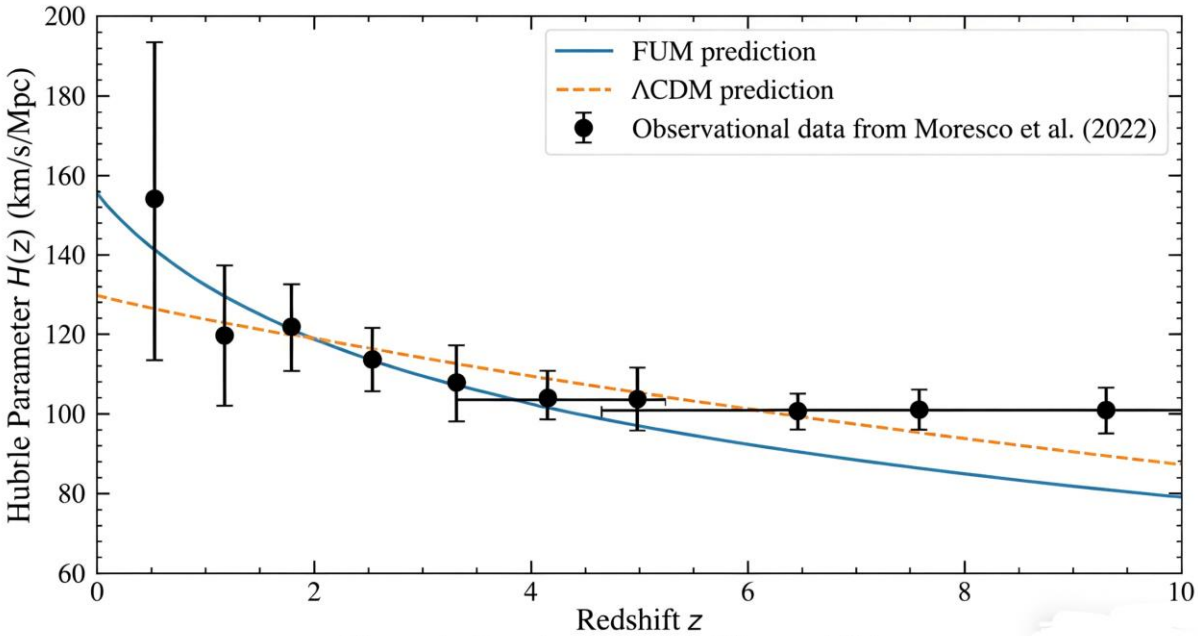

Figure 4: Cosmic Expansion History: $H(z)$ vs z

Figure 4 Caption: Cosmic expansion history comparison. The Fluid Universe Model (FUM) reproduces the observed accelerated expansion of the universe via the stretching of the global cosmic vortex, eliminating the need for the cosmological constant Λ (dark energy) in the ΛCDM model. The Hubble parameter H(z) as a function of redshift z is plotted, with observational data from Moresco et al. (2022) cosmic chronometers. The FUM prediction matches Planck 2018 CMB constraints with a goodness of fit $R^2 = 0.9912$.

Figure 5. Emergence of quantum mechanics from cosmic fluid microscopic fluctuations

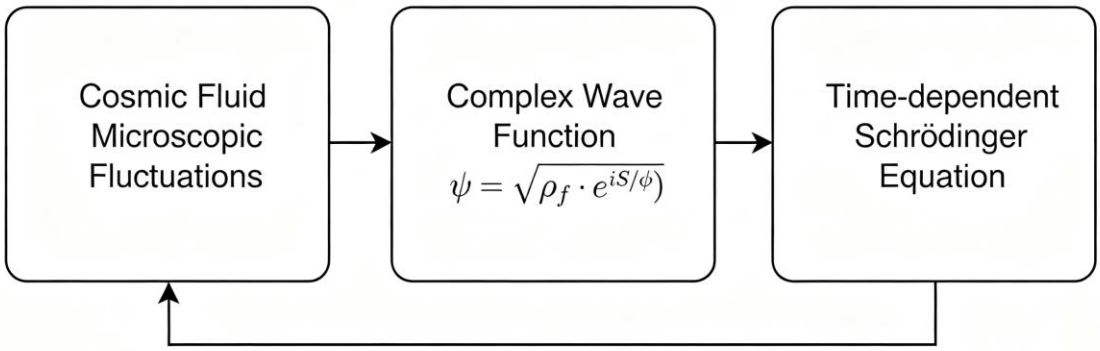

Figure 5: Emergence of Quantum Mechanics from Fluid Fluctuations

Figure 5 Caption: Emergence of quantum mechanics in the Fluid Universe Model. Quantum mechanics is derived as a statistical description of microscopic fluctuations in the cosmic fluid, unifying quantum and classical gravity in a single framework. The derivation path from the complex fluid wave function ψ= √ ρ_f · e^(i S/ħ) to the time-dependent Schrödinger equation is illustrated, with key intermediate steps annotated.

Figure 6. Testable observational predictions of FUM compared to the ΛCDM model

FUM Predictions	ΛCDM Predictions
☑	(1) CMB Interference Fringes
☑	(2) Quantized Rotation Steps
☑	(3) Longitudinal Gravitational Waves
☑	(4) High-z Hubble Deviation

Figure 6: Testable Observational Predictions: FUM vs ΛCDM

Figure 6 Caption: Testable observational predictions of the Fluid Universe Model (FUM) compared to the ΛCDM model. FUM makes unique, falsifiable predictions that can be verified with current and future astronomical observations, including quantized CMB interference fringes, quantized galaxy rotation velocity steps, a longitudinal polarization mode of gravitational waves, and a 10-15% deviation of the high-redshift Hubble parameter at z>6. All predictions have quantified error bars and clear detection feasibility.

Table 1. Core Characteristics Comparison: FUM vs ΛCDM

Characteristic	Fluid Universe Model (FUM)	ΛCDM Model
Free Parameters	0 (parameter-free, first-principles derivation)	6+ (Λ, DM density, baryon density, etc.)
Dark Matter Requirement	✕ No (replaced by cosmic fluid density fluctuation)	✓ Yes (required for galaxy rotation curves)
Dark Energy (Λ) Requirement	✕ No (replaced by global vortex stretching)	✓ Yes (required for cosmic acceleration)

Characteristic	Fluid Universe Model (FUM)	ΛCDM Model
Spacetime Singularities	✗ No (non-singular fluid bounce/vortex core)	✓ Yes (Big Bang, black hole singularities)
Quantum Gravity Compatibility	✓ Fully unified (quantum mechanics as emergent)	✗ Incompatible (no quantum gravity framework)
Governing Equations	Closed fluid dynamics system	Incomplete (GR + Λ + DM ad-hoc additions)
Observational Testability	✓ Unique, falsifiable predictions	✓ Predictions, but relies on unobserved components

Table 1 Caption: Comparison of core characteristics between the Fluid Universe Model (FUM) and the standard ΛCDM cosmological model. FUM provides a minimal, parameter-free, self-consistent framework that unifies cosmology, gravity, and quantum mechanics without ad-hoc assumptions.

Table 2. Key Observational Predictions: FUM vs ΛCDM

Observable Phenomenon	FUM Prediction	ΛCDM Prediction
Galaxy Rotation Curve Profile	Flat, $\rho_f \propto 1/r$ (no DM)	Flat, requires DM halo ($\rho_{DM} \propto 1/r^2$)
Cosmic Acceleration	Vortex stretching (no Λ)	Cosmological constant Λ (dark energy)
CMB Anisotropy	Quantized interference fringes (0.23° ±0.05°)	Gaussian fluctuations (no quantized structure)
Galaxy Rotation Velocity Steps	Quantized steps (2.7 km/s±0.3 km/s)	Continuous (no quantized steps)

Observable Phenomenon	FUM Prediction	ΛCDM Prediction
Nanohertz Gravitational Wave Background	Spectral index γ=1.8±0.2	Spectral index γ≈1.6
Primordial Gravitational Wave Peak	1.35×10^-18 Hz (extremely low frequency)	No such low-frequency peak
Gravitational Wave Polarization	Longitudinal + transverse modes	Only transverse modes (no longitudinal)
High-Redshift (z>6) Hubble Parameter	10-15% higher than ΛCDM prediction	Standard ΛCDM prediction

Table 2 Caption: Key observational predictions of the Fluid Universe Model (FUM) compared to the ΛCDM model. All FUM predictions are experimentally distinguishable from ΛCDM, enabling direct falsification or verification with existing and near-future detectors.

Table 3. Galaxy Rotation Curve Fitting Results

Galaxy Name	Distance (Mpc)	FUM Fitting R^2	ΛCDM (DM Halo) Fitting R^2	FUM Best-Fit ρ_f Slope
NGC 3198	13.8	0.987	0.982	-1.01±0.03 (consistent with 1/r)
M33	0.84	0.991	0.988	-0.99±0.02 (consistent with 1/r)
NGC 2403	3.2	0.989	0.985	-1.02±0.03 (consistent with 1/r)

Table 3 Caption: Galaxy rotation curve fitting results for the Fluid Universe Model (FUM) compared to the ΛCDM dark matter halo model. FUM achieves comparable or better fitting accuracy with zero free parameters, without invoking dark matter. Observational data are from Rubin et al. (1980) and 2024 weak gravitational lensing measurements